

ACHINTYA NARULA

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MACHINE LEARNING | DATA SCIENCE | PYTHON | SQL | NLP

PROFILE

Final-year B.Tech CSE student with an AI & ML minor focused on applied machine learning, NLP, and data systems. Built end-to-end ML pipelines with leakage-safe preprocessing, cross-validation, hyperparameter tuning, explainability, API inference, and dimensional ETL/data-warehouse workflows. Targeting ML engineering, data science, and applied AI/data roles.

EDUCATION

Shaheed Bhagat Singh State University

Expected May 2027

B.Tech, Computer Science and Engineering | Minor: Artificial Intelligence & Machine Learning

CGPA: 7.53/10 | Relevant Coursework: Artificial Intelligence, Machine Learning, Data Mining & Data Warehousing, DBMS, Algorithms

AI / ML & DATA SKILLS

Machine Learning: scikit-learn, XGBoost, Logistic Regression, Random Forest, classification, preprocessing pipelines, feature engineering, 5-fold stratified cross-validation, GridSearchCV

Evaluation & Explainability: ROC-AUC, PR-AUC, precision, recall, F1, confusion matrices, out-of-fold threshold selection, SHAP, permutation importance, leakage prevention

Python & Data: Python, pandas, NumPy, SQL, T-SQL, Power BI, Joblib; data cleaning, EDA, preprocessing, structured-data workflows, model serialization

Data Engineering & Analytics: SQL Server, ETL, staging, star schema, fact/dimension modeling, surrogate keys, Date Dimension, SCD Type 1 MERGE, CTEs, window functions

NLP & Deployment: TF-IDF, word/bigram features, text classification, FastAPI, Docker, pytest, REST/JSON, Git/GitHub, GitHub Actions

SELECTED MACHINE LEARNING & DATA PROJECTS

Customer Churn Prediction & Explainability | Python, scikit-learn, XGBoost, SHAP

Sep 2026

- Built an end-to-end ML pipeline on the 7,043-row IBM Telco Customer Churn sample dataset using leakage-safe preprocessing and training-only 5-fold stratified GridSearchCV across Logistic Regression, Random Forest, and XGBoost.
- Selected XGBoost by cross-validated ROC-AUC and achieved 0.8486 ROC-AUC and 0.6585 PR-AUC on an untouched 25% holdout; analyzed precision, recall, F1, and confusion-matrix trade-offs.
- Added out-of-fold threshold optimization, SHAP and permutation explainability, Joblib persistence, FastAPI inference, Docker-ready serving, automated tests, and GitHub Actions CI.

Sales Analytics & Data Warehouse Pipeline | SQL Server, T-SQL, Python, pandas

Sep 2026

- Designed an OLTP-to-warehouse flow with staging, FactSales, Customer/Product/Date dimensions, surrogate keys, and SCD Type 1 MERGE ETL in SQL Server.
- Generated and validated 500 customers, 60 products, and 5,000 sales rows with Python/pandas quality checks; built analytical SQL using joins, CTEs, LAG, ROW_NUMBER, Power BI-ready metrics, and automated Python/SQL contract checks in CI.

IssueSense | NLP Text Classification | Python, pandas, scikit-learn, TF-IDF

Sep 2026

- Built a reproducible bug/feature/documentation classifier using TF-IDF word/bigram features and Logistic Regression; achieved 82.6% accuracy and 82.7% macro-F1 on a stratified holdout.
- Added per-class metrics, confusion-matrix analysis, confidence-based review routing, Joblib persistence, JSON reports, dataset validation, and 15 automated tests with GitHub Actions CI.

ADDITIONAL EXPERIENCE

Full-Stack Development Intern | Solitaire Infosys

Jul 2025 - Aug 2025

- Built application features and API/data integrations with React, Node.js, Express, and MongoDB in a collaborative development environment; received the Best Intern Award.

Web Development Intern | ShadowFox

May 2025 - Jun 2025

- Completed REST API integration, testing, and Git-based development tasks, strengthening practical programming and data-flow fundamentals.

CERTIFICATIONS & LEADERSHIP

Python for Everybody Specialization - University of Michigan | Crash Course on Python - Google | Power BI Fundamentals

Technical Head, GDG On Campus, SBSSU - organize developer workshops, hackathons, hands-on GenAI labs, and peer-learning activities.